Outline

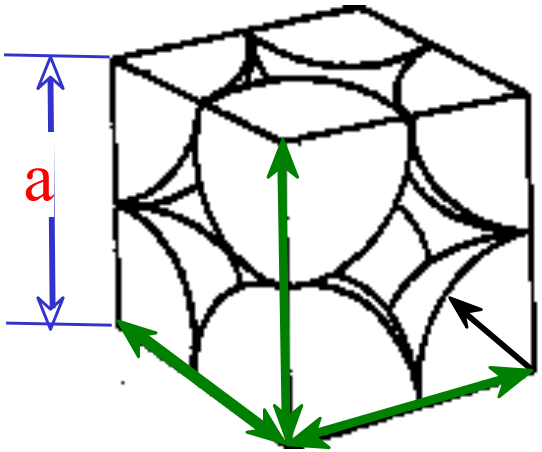
- Review Crystal structure
- Stacking sequence
- theoretical Density
- Crystallographic Directions & Planes
  - example

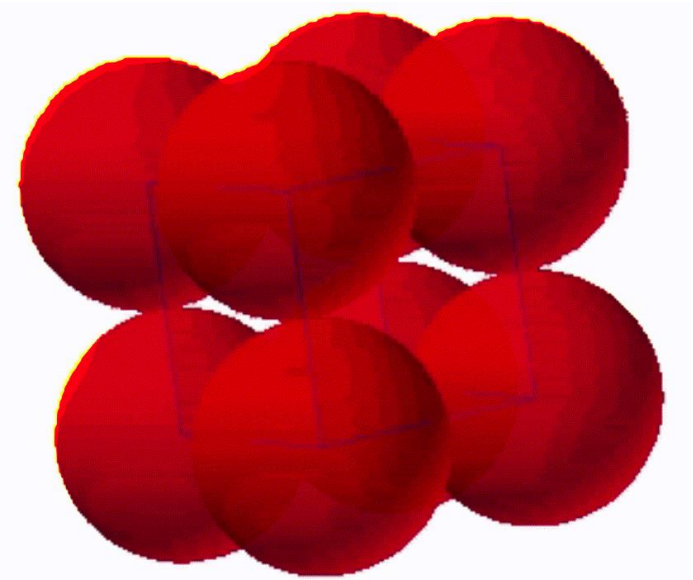
Outline:

- Review of Crystal structure
- Stacking Sequence
- Theoretical Density
- Crystallographic Directions and Planes
  - *examples*

Review: *Crystal Structure*

- Close-packed directions are ..cube..edges



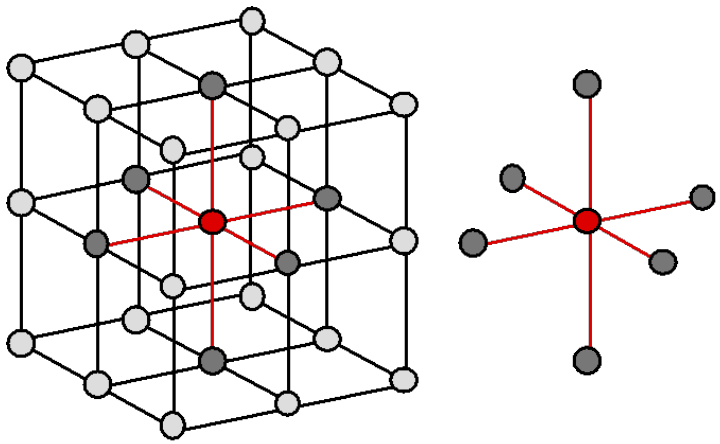


- Contains = ..1 atom/unit cell...
- Coordination # = ..6.. *(# nearest neighbors)*

This is a ..SC.. unit cell

Simple cubic

$$APF = \frac{1 \cdot \frac{4}{3}\pi (0.5a)^3}{a^3}$$

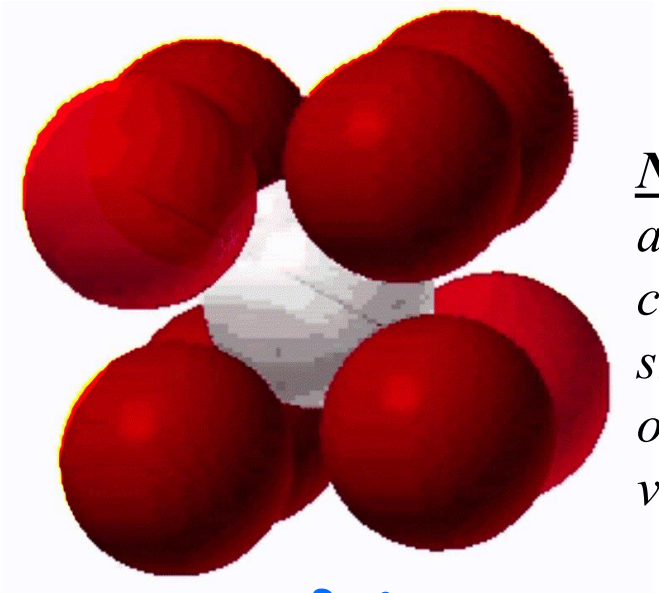


- APF for a simple cubic structure = ...

Review: *Crystal Structure*

- Close packed directions are ……Cube diagonals……

- Coordination # = ……



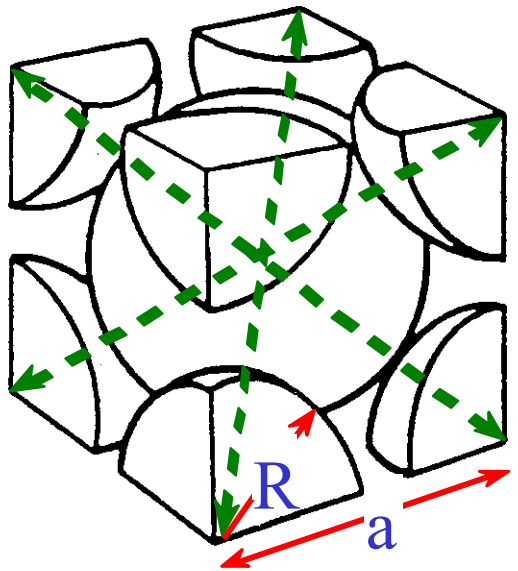
Note: *All atoms are identical; the center atom is shaded differently only for ease of viewing.*

This is a ……BCC…… unit cell

Body centric

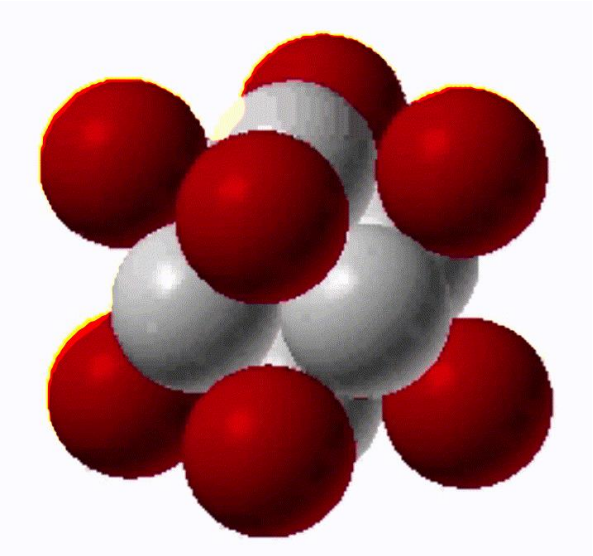
$$APF = \frac{2\left(\frac{4}{3}\pi\left(\sqrt{3}a/4\right)^3\right)}{a^3}$$

- APF for a BCC structure = …….



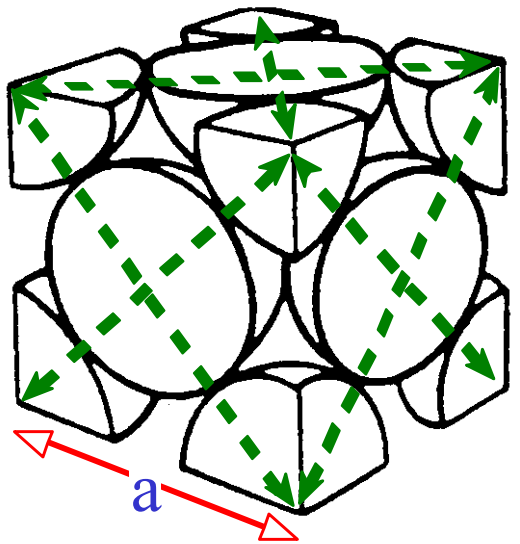
Review: *Crystal Structure*

- Close packed directions are ……face diagonals……
- Coordination # = ……12……



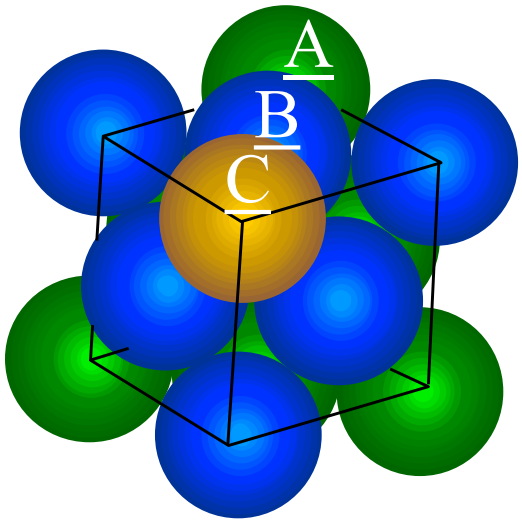
This is a ……FCC…… unit cell

$$APF = \frac{4\left(\frac{4}{3}\pi\left(\sqrt{2}a/4\right)^3\right)}{a^3}$$



- APF for a FCC structure = ……0.74……

FCC Stacking Sequence



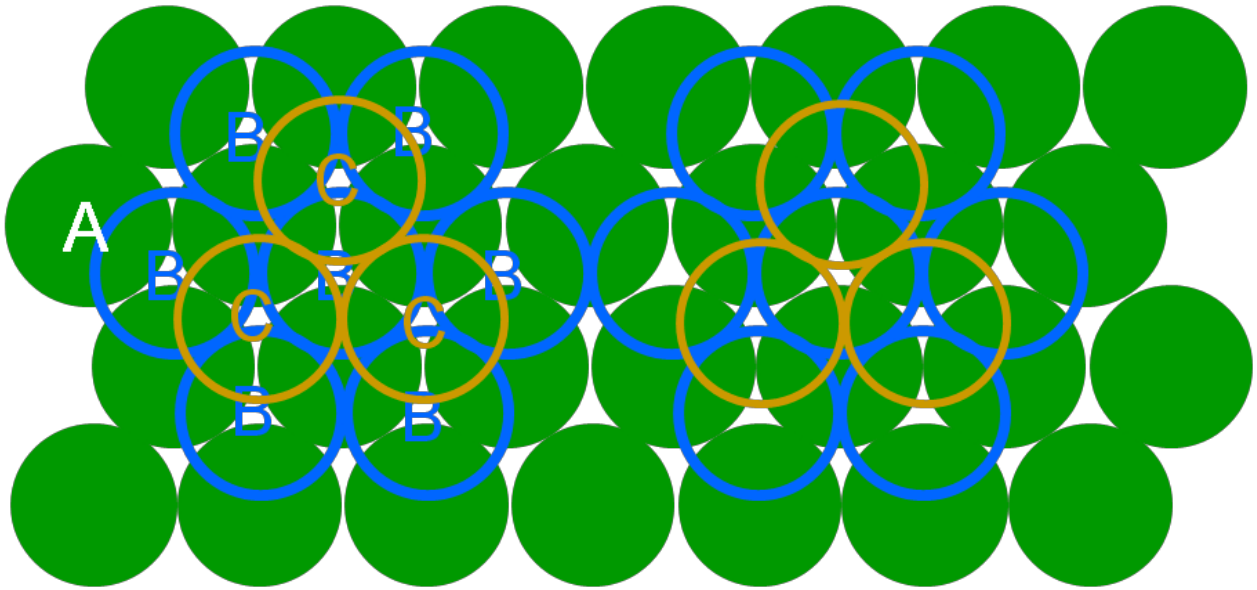
- FCC Unit Cell

3 layers

A sites

B sites

C sites



2D Projection

- ABCABC...etc Stacking Sequence

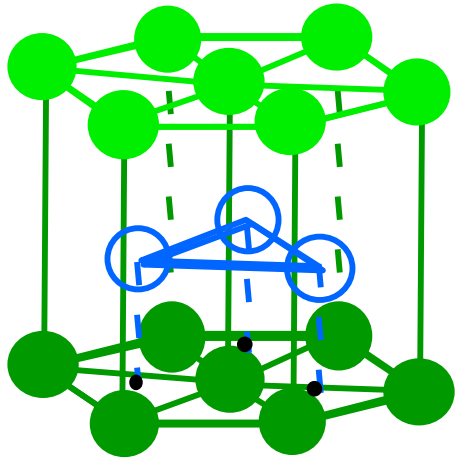
Review: *Crystal Structure*

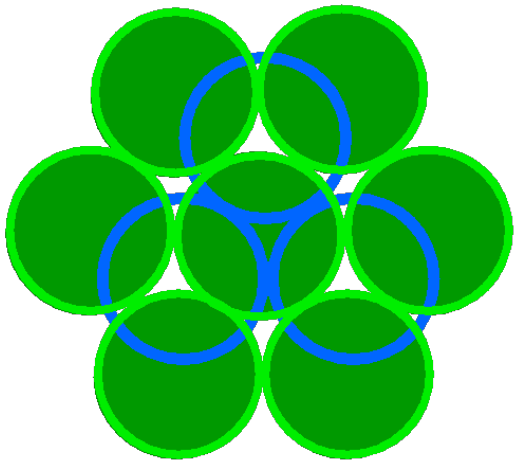
- Coordination # = 12
- APF = 0.74

A sites

B sites

A sites





2D Projection

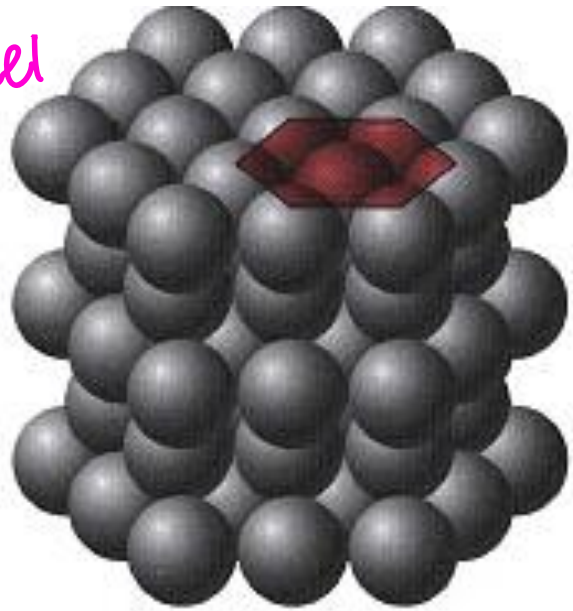
Top layer

Middle layer

Bottom layer

parallel

This is a HCP unit cell

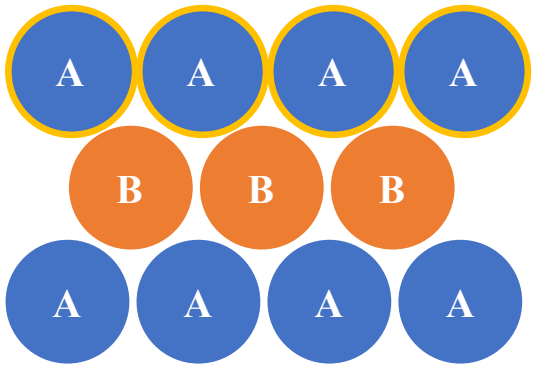


- Stacking Sequence: ABA

HCP vs. FCC: *the Third Layer Decides*

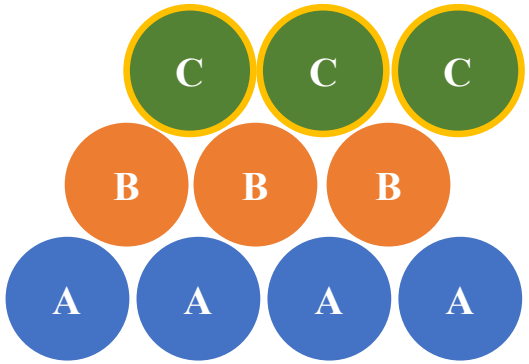
Both start the same way: a close-packed A layer, then a B layer in the hollows.

HCP — ABAB



Third layer sits directly above the A atoms. The C hollows stay empty, so the pattern repeats every two layers: ABAB…

FCC — ABCABC



Third layer shifts into the empty C hollows — a new position. Only the fourth layer returns to A: ABCABC…

Same packing, different order: CN = 12 and APF = 0.74 for both. The stacking sequence is what differs — and it is why FCC and HCP deform so differently.

Example Problem

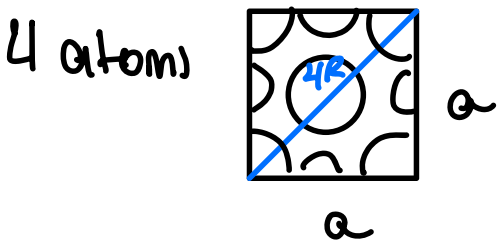
If you know

- the crystal structure,
- the atomic radius
- the atomic weight,

you can calculate the density of a particular material

Example:

Copper has an atomic radius 0.128 nm an FCC crystal structure and an atomic weight of 63.5 g/mol. Calculate its density.



$N_A = 6.02 \times 10^{23}$ atoms/mol

$a = \sqrt{8}R$

$V = a^3 = 0.04745\ nm^3$

$\rho = \frac{n}{V} = \frac{4.2172 \times 10^{-22}\ g}{0.04745 \times 10^{-21}\ cm^3} = 8.8876 \frac{g}{cm^3}$

Crystallographic Directions, and Planes

Now that we know how atoms arrange themselves to form crystals, *we need a way to identify directions and planes of atoms*.

Why?

✓ Deformation under loading (سلب.) occurs on certain crystalline planes and in certain crystallographic directions.

✓ Before we can predict how materials fail, we need to know what modes of failure are more likely to occur.

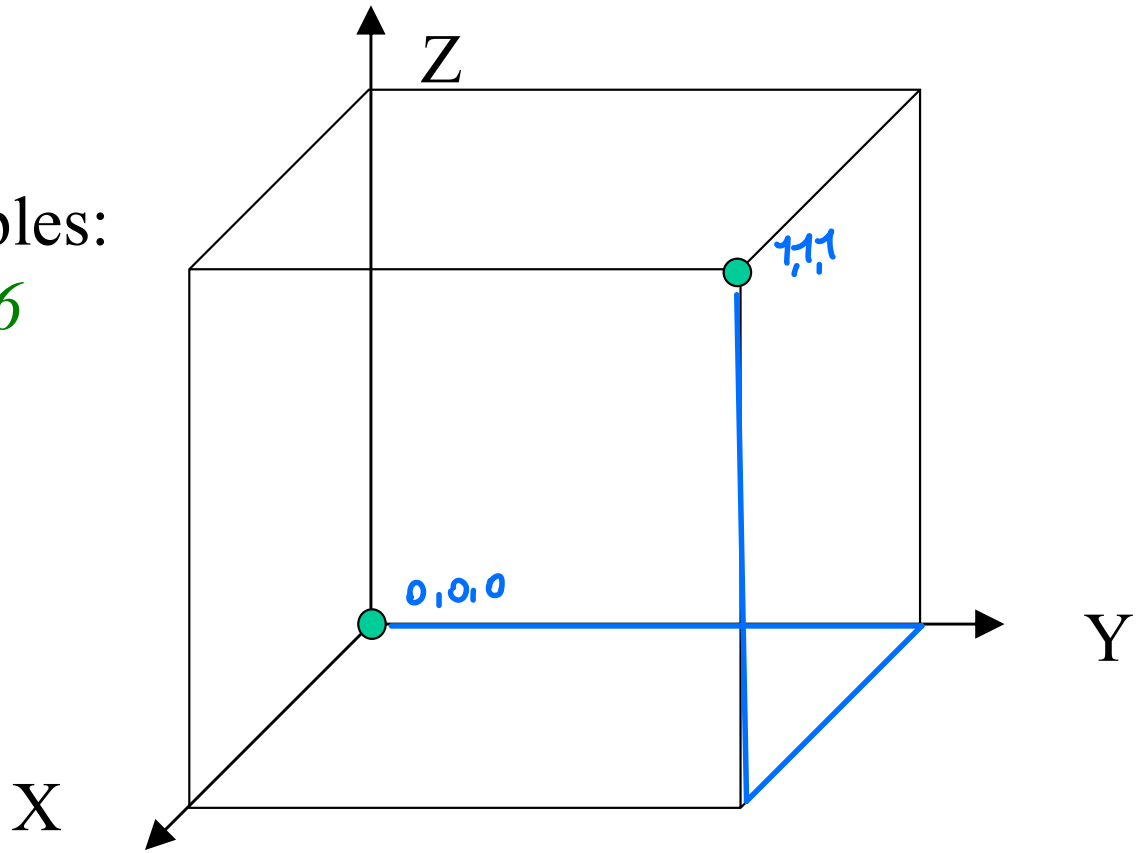
✓ Other properties of materials (*electrical conductivity, thermal conductivity, elastic modulus*) can vary in a crystal with orientation.

Directions and planes are described using three integers - Miller… Indices

Point Coordinates

Point position specified in terms of its coordinates as fractional multiples of the unit cell edge lengths

See examples:
3.5 and 3.6



General Rules for Lattice Directions, Planes & Miller Indices

- Miller indices used to express lattice *planes* and *directions*
- x, y, z are the axes (on arbitrarily positioned origin)
  - *in some crystal systems these are not mutually ⊥*
- a, b, c are lattice parameters (*length of unit cell along a, b, c sides*)
- h, k, l are the Miller indices for planes and directions - expressed as planes: (hkl) and directions: [hkl]

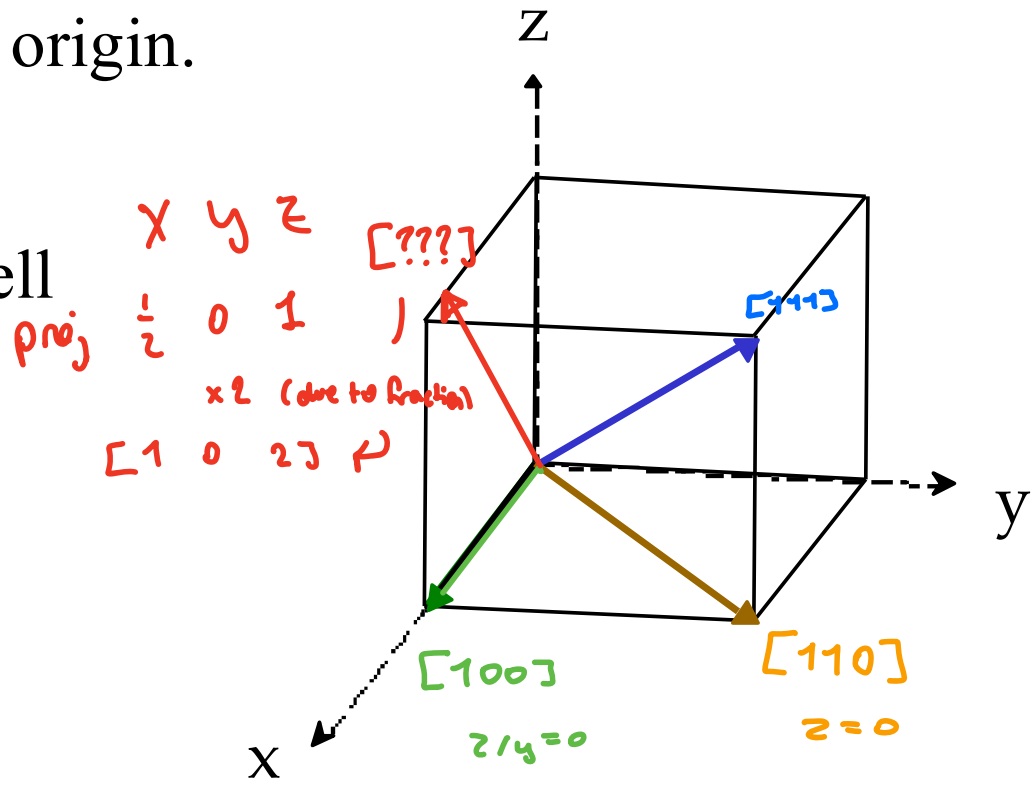
- Conventions for naming
  - There are NO COMMAS between numbers
  - Negative values are expressed with a bar over the number
    - *Example: -2 is expressed* $\bar{2}$

- Crystallographic direction:
  - [123]
  - [100]
  - … etc.

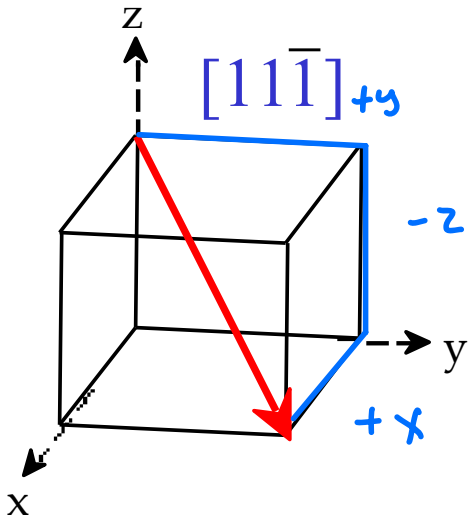
Miller Indices for Directions

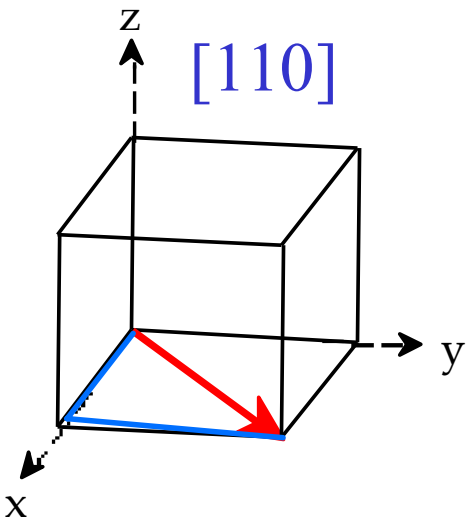
Method

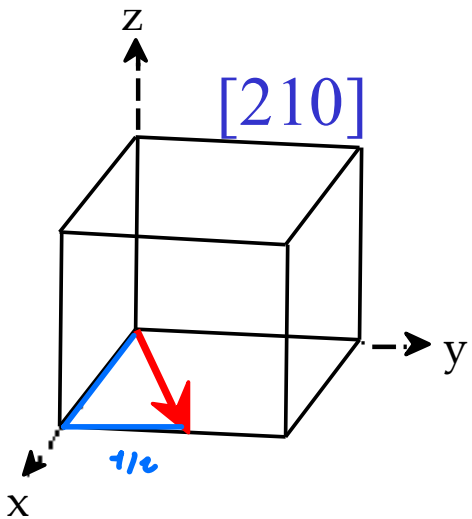
- Draw vector, define tail as origin.
- Determine length of the vector projection in unit cell dimensions, a, b, and c
- Remove fractions by multiplying by smallest possible factor
- Enclose in square brackets

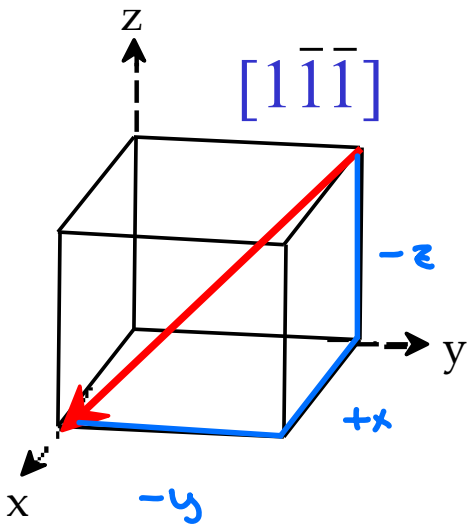


Example - Naming Directions





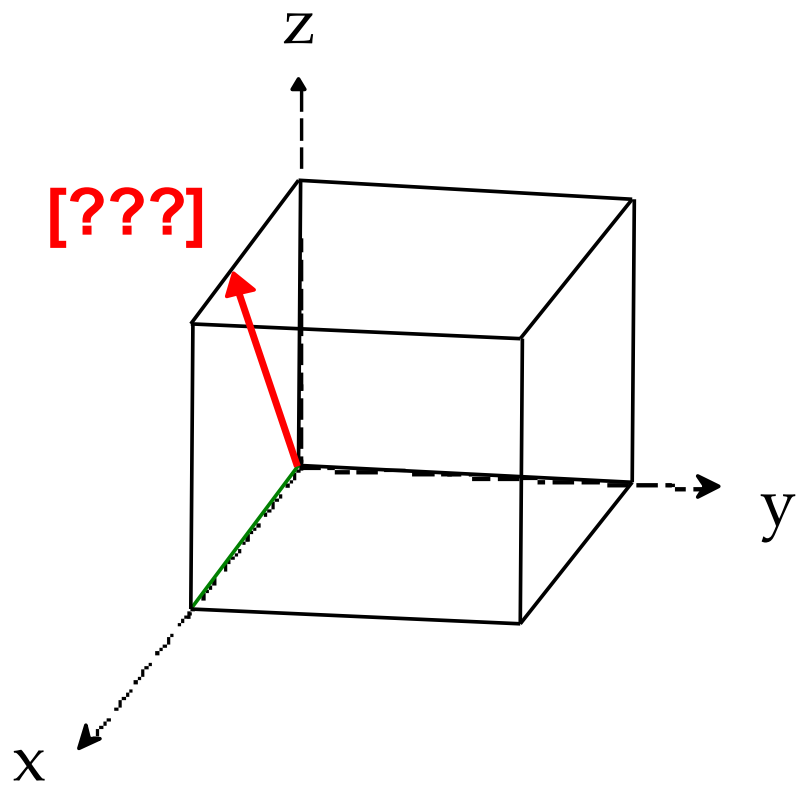




Miller Indices for Directions

Use the QR code shown in the classroom. This one won't work.

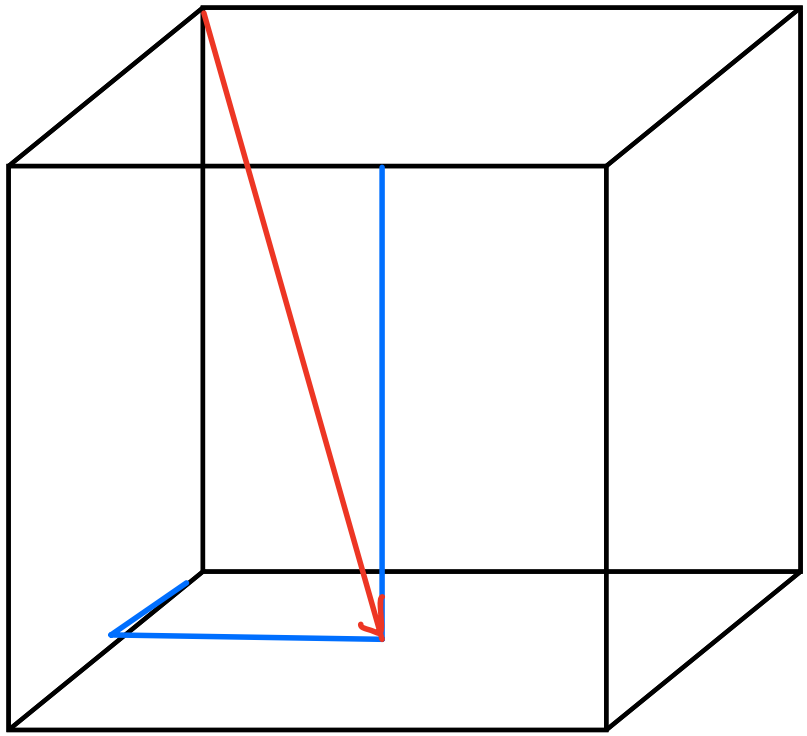
What is [? ? ?]



Example - Drawing Directions

- Draw $[11\bar{2}]$ and $[\bar{1}\bar{1}\bar{1}]$

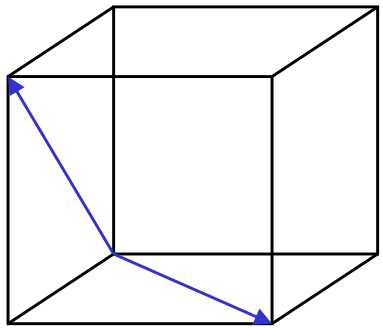
$\frac{1}{2}\ \frac{1}{2}\ -1$



The HW problems will give you more practice with directions

Families of Directions

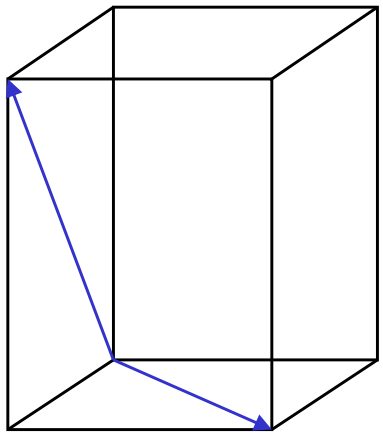
- Equivalence of directions



Only true for cubic

[101] = [110]

cubic



[101] ≠ [110]

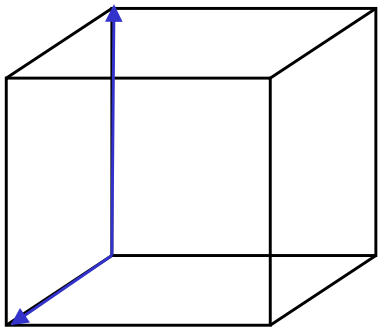
tetragonal

- <123> Family of directions
  - [123], [213], [312], [132], [231], [321]
    - *only in a cubic crystal*

for cubic, any coords containing 1, 2 and 3 are eq regardless of order/sign

*In the cubic system directions having the same indices regardless of **order** or **sign** are equivalent.*

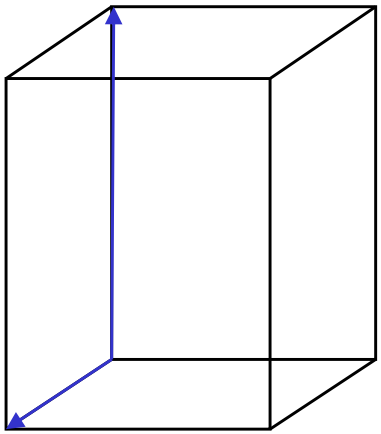
Isotropy vs. Anisotropy in Single Crystals



[100] = [001]

cubic

Isotropic



[100] ≠ [001]

tetragonal

Anisotropic

- Properties are independent of direction → isotropic material
- Directionality of properties → anisotropic material

Miller Indices for Planes

brackets or imp [] = directions

- (hkl) Crystallographic plane
- {hkl} Family of crystallographic planes
  - e.g. (hkl), (lhk), (hlk) … etc.

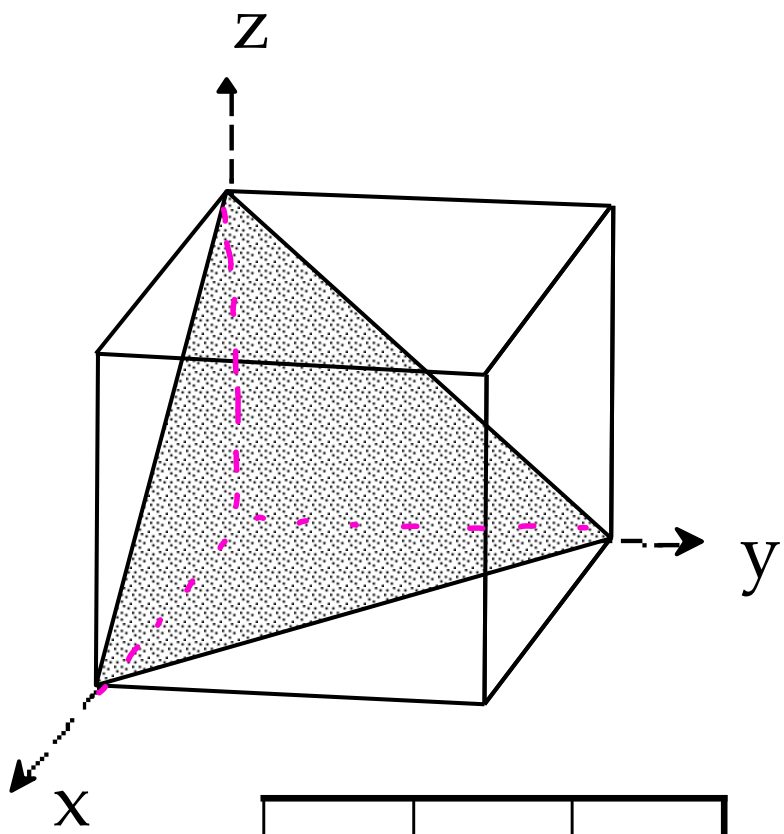
*In the cubic system planes having the same indices regardless of **order** or **sign** are equivalent*

- Hexagonal crystals can be expressed in a four index system (u v t w)
  - *Can be converted to a three index system using formulas*

Miller Indices for ***PLANES***

Method

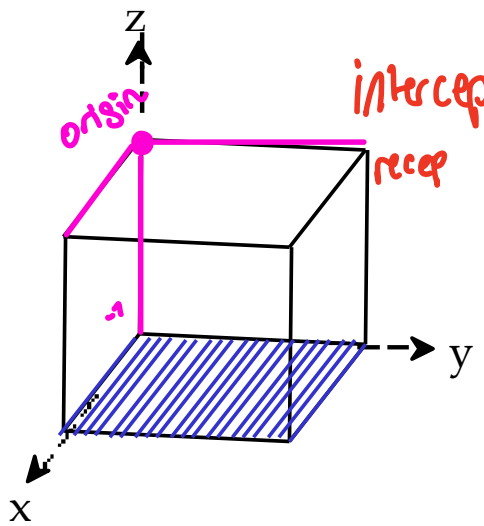
- If the plane passes through the origin, select an equivalent plane or move the origin
- Determine the intersection of the plane with the axes in terms of a, b, and c
- Take the reciprocal *(1/∞ = 0)*
- Convert to smallest integers *(optional)*
- Enclose by parentheses



	x	y	z
Intercepts	1	1	1
Reciprocals	1	1	1

see example 3.10

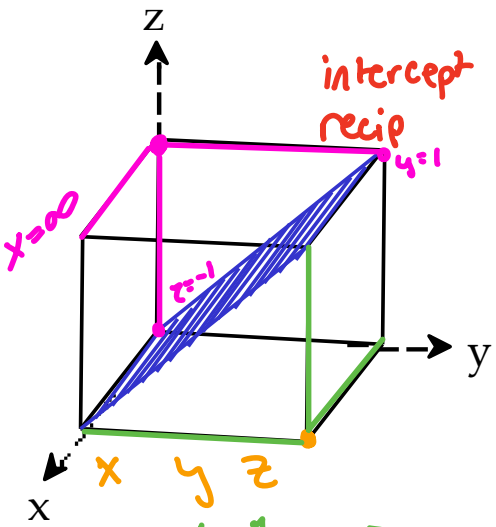
Crystallographic Planes



∅ intercept

	x	y	z
intercept	∞	∞	-1
recip	0	0	-1

$(0\,0\,\bar{1})$



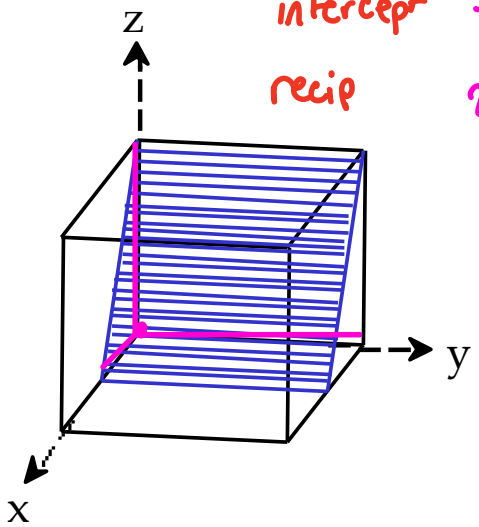
	x	y	z
intercept	∞	1	-1
recip	0	1	-1

$(0\,1\,\bar{1})$

equivalent

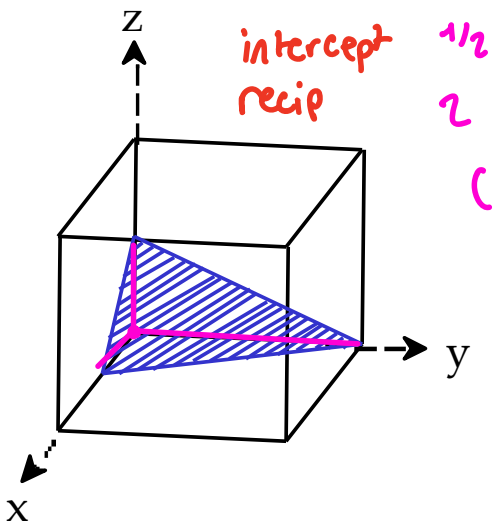
	x	y	z
intercept	∞	-1	1
recip	0	-1	1

$(0\bar{1}1)$



	x	y	z
intercept	1/2	∞	1
recip	2	0	1

$(2\,0\,1)$



	x	y	z
intercept	1/2	1	1/2
recip	2	1	2

$(2\,1\,2)$

Next time:

Linear and Planner Densities